

GL521MG @Track Air Interface Protocol –Stolen Vehicle Recovery

EGPRS/LTE Cat-M1/LTE Cat-NB2/GNSS Tracker

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0. Revision History

Version	Date	Author	Description of Change
1.00	2024-06-20	Cooper Wu	Initial

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1. Overview

This document is a complement of the @Track Air Interface Protocol for Stolen Vehicle Recovery function. This document describes how to implement the Stolen Vehicle Recovery function based on the @Track Air Interface Protocol.

Please refer to @Track Air Interface Protocol document for its detailed information.

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2. Stolen Vehicle Recovery

2.1. Command And Acknowledgement

2.1.1. Stolen Vehicle Recovery

Small in size, the device (Ghost Device) can be hidden in many places in the vehicle. Another device (Wired Device) should be installed with permanent 12V power source. Both devices support Bluetooth Low Energy (BLE) connectivity and can generate the **+RESP:GTSVR** alert message if they determine a theft has occurred and the other one has been removed.

➤ AT+GTSVR=

Example: AT+GTSVR=gl521m,1,111122223333,10,1,865585040008967,1,,,,,FFFF\$			
Parameter	Length (Byte)	Range/Format	Default
Password	4 – 20	'0' – '9' 'a' – 'z' 'A' – 'Z'	gl521m
Mode	1	0 1	0
Target MAC Address	12	000000000000 ~ FFFFFFFF	
Connect Interval	4	10 20 30 60 120 180 240 360 480 720 1440	10
Connect Timeout Count	<=2	1 – 10	1
Target IMEI	15	'0' – '9'	
Continuous Mode Switch	1	0 1	0
Reserved	0		
Reserved	0		
Reserved	0		
Reserved	0		
Serial Number	4	0000 – FFFF	
Tail Character	1	\$	\$

- ✧ <Mode>: The working mode of the stolen vehicle recovery function.
 - 0: Disable this function.
 - 1: Enable this function.
- ✧ <Target MAC Address>: The MAC address of the target device.
- ✧ <Connect Interval>: It defines the interval for connecting to the target device. Unit: minute.
- ✧ <Connect Timeout Count>: A numeral to indicate the number of times the device's Bluetooth cannot find the target device. If the number of connection failures exceeds the threshold specified by this parameter, the device will consider the target device has been removed and report the **+RESP:GTSVR** message.
- ✧ <Target IMEI>: The IMEI of the target device.
- ✧ <Continuous Mode Switch>: It specifies whether the device switches to continuous mode

when the target device is lost or restored.

- 0: Do not switch to continuous mode.
- 1: Change the current working mode to continuous mode when the target device is lost, and reset the working mode to the mode configured by *<Mode Selection>* in @Track Air Interface Protocol when the target device is restored.

The acknowledgment message of the **AT+GTSVR** command:

➤ **+ACK:GTSVR,**

Example: +ACK:GTSVR,550200,869406800001101,GL521MG,0017,20190410103044,0452\$			
Parameter	Length (Byte)	Range/Format	Default
Protocol Version	6	XX0000 – XXXFFF, X ∈ {'A'-'Z','0'-'9'}	
Unique ID	15	IMEI	
Device Name	≤20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Serial Number	4	0000 – FFFF	
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	0000 – FFFF	
Tail Character	1	\$	\$

2.1.2. Real Time Operation

The command **AT+GTRTO** is used to retrieve information from the terminal or control the terminal to execute certain actions.

➤ **AT+GTRTO=**

Example: AT+GTRTO= gl521m,2,SVR,,,,,FFFF\$ AT+GTRTO=gl521m,2,0000000000200000,,,,,FFFF\$ AT+GTRTO= gl521m,15,,,,,FFFF\$ AT+GTRTO= gl521m,16,,,,,FFFF\$				
SN	Parameter	Length (Byte)	Range/Format	Default
1	Password	4 – 20	'0' – '9', 'a' – 'z', 'A' – 'Z'	gl521m
2	Sub Command	≤2	1 – 1C	
3	AT Command Configuration Mask ATI Mask	3 16 8	"SRI" 0000000000000000 – FFFFFFFFFFFFFFFF 00000000 – FFFFFFFF	
4	Reserved	0		
5	Reserved	0		
6	Reserved	0		
7	Sub Command Parameter	≤1		
8	Serial Number	4	(HEX)	
9	Tail Character	1	\$	\$

✧ *<Sub Command>*: The valid values are 0-1C.

- 15: **QBI**. Get device Bluetooth information only via the ASCII message **+RESP:GTQBI**.

- 16: **RST**. Reset Bluetooth pairing information
- ✧ <AT Command/Configuration Mask/ATI Mask>:
- ✧ <AT Command>: To get a single AT command's configuration when <Sub Command> is set to 2, follow the format in the following example. For example, to get the configuration of AT+GTSVR, please set AT+GTRTO=gl521m,2,SVR,,,,,0015\$, and get it via **+RESP:GTALS**.
- ✧ <Configuration Mask>: If <Sub Command> is set to 2, the configuration information of the specified <Configuration Mask> can be obtained via the message **+RESP:GTALC**. The Configuration Mask must be 16 bytes. If it's less than 16 bytes, '0' will be added in the high bytes of the Configuration Mask.

Configuration Mask Table:

Mask Bit	Item
Bit 22	Reserved
Bit 21	SVR
....	
Bit 0	Reserved

2.2. Stolen Vehicle Recovery Report

2.2.1. Report for Real Time Querying

2.2.1.1. +RESP:GTALS

After the device receives the command **AT+GTRTO** to get sub AT command configuration information, it will send the configuration information to the backend server via the message **+RESP:GTALS**. Configuration information varies with different AT commands. For example, to get SVR configuration, set **AT+GTRTO=gl521m,2,SVR,,,,,FFFF\$**.

➤ +RESP:GTALS,

Example:

+RESP:GTALS,D80101,860112043797648,,SVR,1,B91292E09C20,10,1,015700000034015,0,,,,,20200916064207,0090\$

Parameter	Length (Byte)	Range/Format	Default
Protocol Version	6	XX0000 – XXXFFF, X ∈ {'A' – 'Z', '0' – '9'}	
Unique ID	15	IMEI	
Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' ' ' ' _ '	
Sub AT Command	3	'a' – 'z' 'A' – 'Z' ' '	
Mode	1	0 1	0
Target MAC Address	12	000000000000 ~ FFFFFFFF	
Connect Interval	4	10 20 30 60 120 180 240 360 480 720 1440	10
Connect Timeout Count	<=2	1 – 10	1
Target IMEI	15	'0' – '9'	
Continuous Mode Switch	1	0 1	0
Reserved	0		
Reserved	0		
Reserved	0		
Reserved	0		
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	0000 – FFFF	
Tail Character	1	\$	\$

2.2.1.2. +RESP:GTALC

After the device receives the command **AT+GTRTO** to read the configurations, it will send corresponding configuration information to the backend server via the message **+RESP:GTALC** according to the configuration mask. The **+RESP:GTALC** does not support HEX format report.

➤ **+RESP:GTALC,**

Example: +RESP:GTALC,D80101,015425000021222,GL521M,0000000000000403,BSI,4gnet,,,2gnet,,,3,0,SR I,1,,1,218.17.46.11,65535,218.17.46.11,65533,18565713530,0,0,0,20,0,,20200117173015,0199\$			
Parameter	Length (Byte)	Range/Format	Default
Protocol Version	6	(HEX)	
Unique ID	15	(IMEI)	
Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Configuration Mask	16	0000000000000000 - FFFFFFFFFFFFFFFF	
SVR	3	SVR	SVR
Mode	1	0 1	0
Target MAC Address	12	000000000000 ~ FFFFFFFFFF	FFFFFFFFF
Connect Interval	4	10 20 30 60 120 180 240 360 480 720 1440	10
Connect Timeout Count	<=2	1 – 10	1
Target IMEI	15	'0' – '9'	0000000000 00000
Continuous Mode Switch	1	0 1	0
Reserved	0		
Reserved	0		
Reserved	0		
Reserved	0		
Count Number	4	0000 – FFFF	
Tail Character	1	\$	\$

2.2.1.3. +RESP:GTSVR

Example: +RESP:GTSVR,D80101,359464038747858,GL521MG,1,2121212121,7700000000001E00820 096,0,0,11.1,346,256.4,117.128163,31.835908,20190409005446,0460,0000,550B,B969,,2019 0409110309,1888\$			
Parameter	Length (Byte)	Range/Format	Default
Protocol Version	6	XX0000 – XXFFFF, X ∈ {'A' – 'Z', '0' – '9'}	
Unique ID	15	IMEI	

Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' ' ' '_'	
SVR Working State	1	0 1 2	
Target MAC Address	12	000000000000 - FFFFFFFF	
SVR Appending Information	22		
Target State	1	0 1	0
GNSS Accuracy	1	0 - 6	
Speed	<=5	0.0 – 999.9 km/h	
Azimuth	<=3	0 – 359	
Altitude	<=8	(-)xxxxx.x m	
Longitude	<=11	-180 – 180	
Latitude	<=10	-90 – 90	
GNSS UTC Time	14	YYYYMMDDHHMMSS	
MCC	0 4	0XXX	
MNC	0 4	0XXX	
LAC	0 4	XXXX	
Cell ID	<=8	XXXX	
Reserved	0		
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	0000 – FFFF	
Tail Character	1	\$	\$

✧ <SVR Working State>: It indicates the status in which the **+RESP:GTSVR** alert message is generated.

- 0: The device loses connection with the target device.
- 1: The device regains connection with the target device.
- 2: The device fails to match the target device during installation phase.

✧ <SVR Appending Information>: It indicates additional Bluetooth information.

Parameter	Length (byte)	Range/Format	Default
SVR Lost Reason	1	3 7 8 9 B	
SVR Trial 1 Result	1	1 – 7	
SVR Trial 2 Result	1	0 – 7	
SVR Trial 3 Result	1	0 – 7	
SVR Trial 4 Result	1	0 – 7	
SVR Trial 5 Result	1	0 – 7	
SVR Trial 6 Result	1	0 – 7	
SVR Trial 7 Result	1	0 – 7	
SVR Trial 8 Result	1	0 – 7	
SVR Trial 9 Result	1	0 – 7	
SVR Connect Duration	4	0000 – 1388	
SVR Read Duration	4	0000 – 0FA0	
SVR Write Duration	4	0000 – 0FA0	

- SVR Lost Reason>: SVR lost reason.
 - ◆ 3: Bluetooth connection failed.
 - ◆ 7: Bluetooth connection is successful.
 - ◆ 8: Failed to decrypt interactive data.
 - ◆ 9: After the interactive data decryption is successful, the data check is incorrect.
 - ◆ B: After the connection is successful, there is no data interaction.
- <SVR Trial 1 Result>: The first Bluetooth connection result
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 2 Result>: The second Bluetooth connection result
 - ◆ 0: Due to the first Bluetooth connection timeout, or the read/write request timeout, restart the Bluetooth directly, and the Bluetooth connection is not actually initiated this time; or the first Bluetooth SVR execution is successful.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 3 Result>: 3rd Bluetooth connection result
 - ◆ 0: Because the first or second Bluetooth connection timed out, or the read/write request timed out, restart the Bluetooth directly, and the Bluetooth connection was not actually initiated this time; or the first or second Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 4 Result>: 4th Bluetooth connection result
 - ◆ 0: The previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.

- ◆ 5: GL521MG read request timed out.
- ◆ 6: GL521MG write request timed out.
- ◆ 7: GL521MG SVR success.
- <SVR Trial 5 Result>: The 5th Bluetooth connection result
 - ◆ 0: Due to the 4th Bluetooth connection timeout, or the read/write request timeout, restart the Bluetooth directly, and the Bluetooth connection is not actually initiated this time; or the previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 6 Result>: 6th Bluetooth connection result
 - ◆ 0: Due to the 5th or 6th Bluetooth connection timeout, or the read/write request timeout, restart the Bluetooth directly, and the Bluetooth connection is not actually initiated this time; or the previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 7 Result>: The 7th Bluetooth connection result
 - ◆ 0: The previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timeout.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Trial 8 Result>: 8th Bluetooth connection result
 - ◆ 0: Due to the 7th Bluetooth connection timeout, or the read/write request timeout, restart the Bluetooth directly, and the Bluetooth connection is not actually initiated this time; or the previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.

- ◆ 4: GL521MG connection timeout.
- ◆ 5: GL521MG read request timed out.
- ◆ 6: GL521MG write request timed out.
- ◆ 7: GL521MG SVR success.
- <SVR Trial 9 Result>: 9th Bluetooth connection result
- 0: Due to the 7th or 8th Bluetooth connection timeout, or the read/write request timeout, restart the Bluetooth directly, and the Bluetooth connection is not actually initiated this time; or the previous Bluetooth SVR has been successfully executed.
 - ◆ 1: GL521MG has initiated a connection.
 - ◆ 2: GL521MG has initiated a read request.
 - ◆ 3: GL521MG has initiated a write request.
 - ◆ 4: GL521MG connection timed out.
 - ◆ 5: GL521MG read request timed out.
 - ◆ 6: GL521MG write request timed out.
 - ◆ 7: GL521MG SVR success.
- <SVR Connect Duration>: The time between the connection request initiated by the Bluetooth and the read request initiated, the maximum is 5000ms.
- <SVR Read/Write Duration>: The time from when Bluetooth initiates a read/write request to its successful execution, the maximum is 4000ms.
- ✧ <Target State>: A numeral to indicate the status of the target device.
 - 0: Normal.
 - 1: Network jamming.

2.2.1.4. +RESP:GTQBI

The report for real time operation of the sub command QBI.

Example:			
Parameter	Length (byte)	Range/Format	Default
Protocol Version	6	XX0000 – XXFFFF, $X \in \{'A' - 'Z', '0' - '9'\}$	
Unique ID	15	IMEI	
Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Local Bluetooth Mac	12	000000000000 - FFFFFFFF	
Reserved	0		
Reserved	0		
Reserved	0		
Reserved	0		
Reserved	0		

Reserved	0		
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	0000 – FFFF	
Tail Character	1	\$	\$

2.2.1.5. +RESP:GTPNA

Example: +RESP:GTPNA,D80101,015425000020117,GL521M,2,20191011160545,0084\$			
Parameter	/	Range/Format	Default
Protocol Version	6	(HEX)	
Unique ID	15	(IMEI)	
Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Power On Type		1 - 16	
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	(HEX)	
Tail Character	1	\$	\$

✧ <Power On Type>: A numeral to indicate how the terminal is activated.

- 1: Movement detected by the terminal.
- 2: Specified time reached.
- 3: Reserved.
- 4: Manual powering on for the first time.
- 5: RTO command or reboot by GTDOG.
- 6: Abnormal power on, for example, the battery is removed and connected again later.
- 7: RTC error.
- 8: FOTA process.
- 9: Temperature sensor alarm event.
- 10-14: Reserved.
- 15: Bluetooth lost.
- 16: Bluetooth connected.

3.3.3.5 +RESP:GTVR

After the device receives the command **AT+GTRTO** to get the versions (including firmware version, hardware version and module version), it will send the version information to the backend server by the message **+RESP:GTVR**.

Example: +RESP:GTVR,D80101,015425000020117,GL521M,0902,0106,,BG96,BG96MAR02A07M1G,20191011161008,008B\$			
Parameter	Length	Range/Format	Default

Protocol Version	6	(HEX)	
Unique ID	15	(IMEI)	
Device Name	<=20	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Device type	<=10	'0' – '9' 'a' – 'z' 'A' – 'Z' '-' '_'	
Firmware Version	4	(HEX)	
Hardware Version	4	(HEX)	
Bluetooth Version	4	(HEX)	
Modem Hardware Version	<=20		
Modem Software Version	<=50		
Send Time	14	YYYYMMDDHHMMSS	
Count Number	4	(HEX)	
Tail Character	1	\$	\$

- ✧ <Device Type>: The type of the device.
- ✧ <Firmware Version>: The firmware version of the device. The first two characters indicate the major version and the last two characters indicate the minor version. For example, 010A means version 1.10.
- ✧ <Hardware Version>: The hardware version of the device. The first two characters indicate the major version and the last two characters indicate the minor version. For example, 010A means version 1.10.
- ✧ < Bluetooth Version >: The Bluetooth version. The first two characters represent the major version and the last two characters represent the minor version. For example: 010A means the version 1.10.
- ✧ <Modem Hardware Version>: It gives the modem hardware information of this device.
- ✧ <Modem Software Version>: It gives the modem software version information of this device.

Appendix: Message Index

- ✧ Configure Command

[AT+GTSVR](#)

- ✧ Release Command

[AT+GTRTO](#)

- ✧ Event Report

[+RESP:GTALS](#)

[+RESP:GTALC](#)

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[+RESP:GTQBI](#)

[+RESP:GTPNA](#)